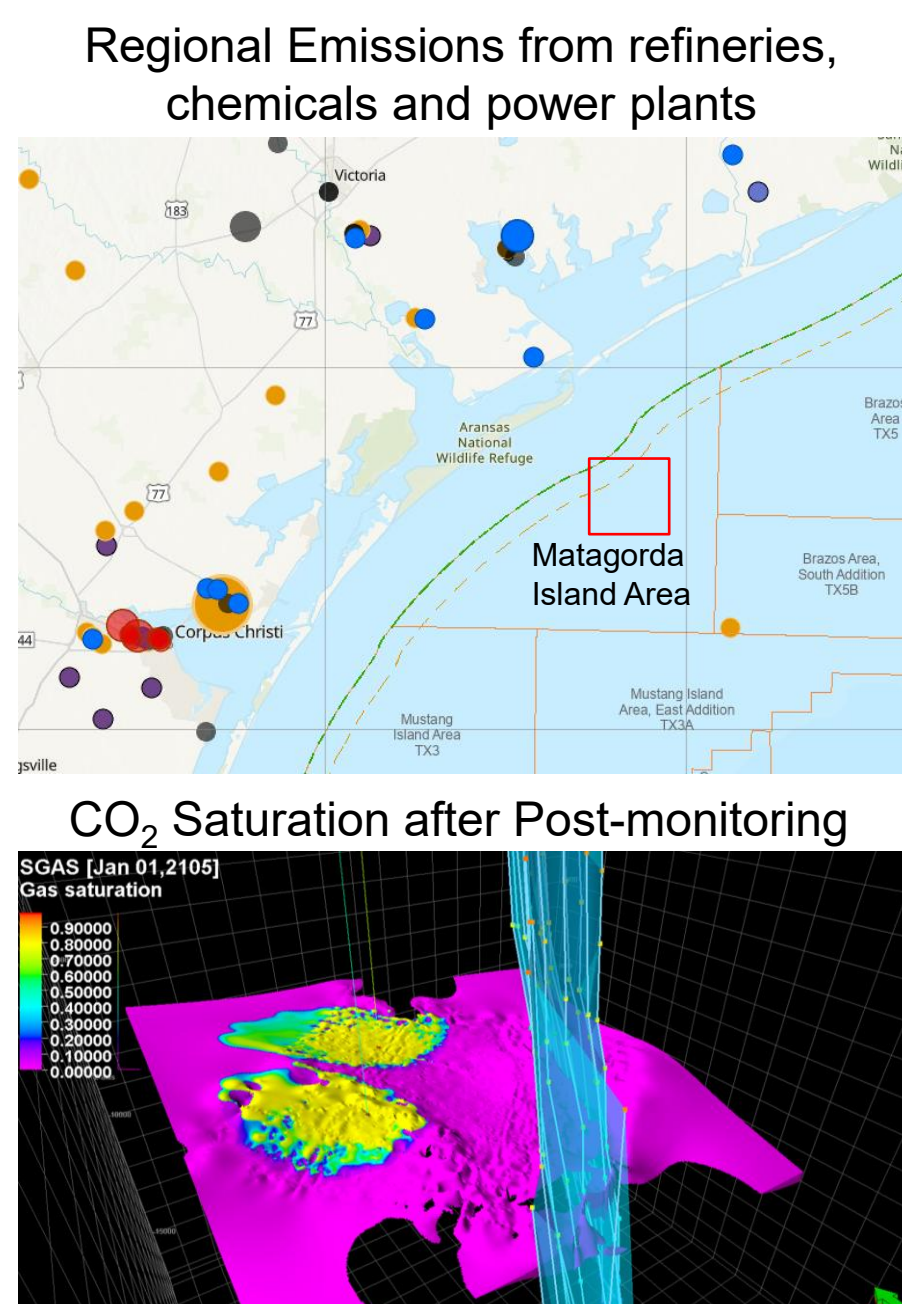


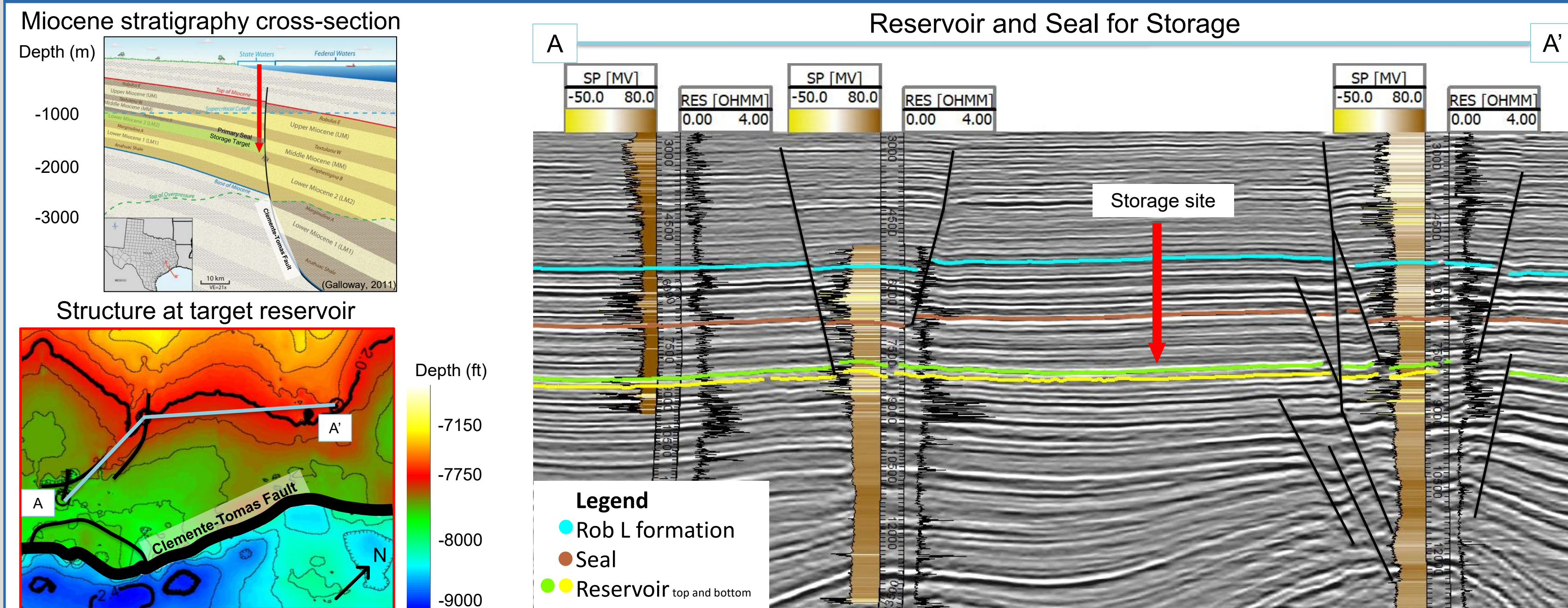
1 Executive Summary

Many CCS projects are being considered in the GoM. We demonstrate the potential of the federal Matagorda Island lease blocks to decarbonize the metropolitan areas of Corpus Christi and Victoria, TX.

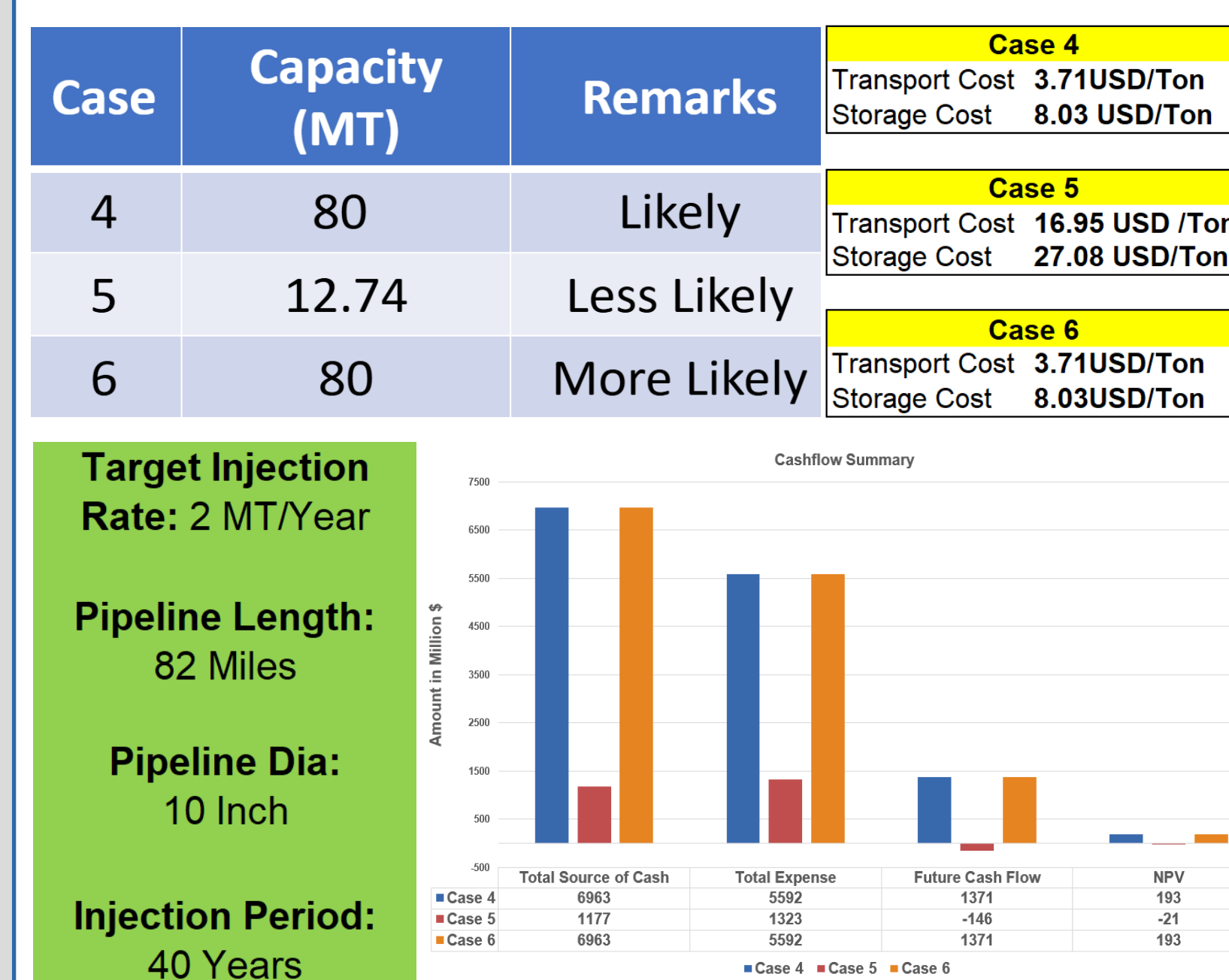
Storage:
80 MT total amount of CO₂
Pipeline length: 82 miles
Economics:
\$5.5B Project Cost
20.52% Rate of Return



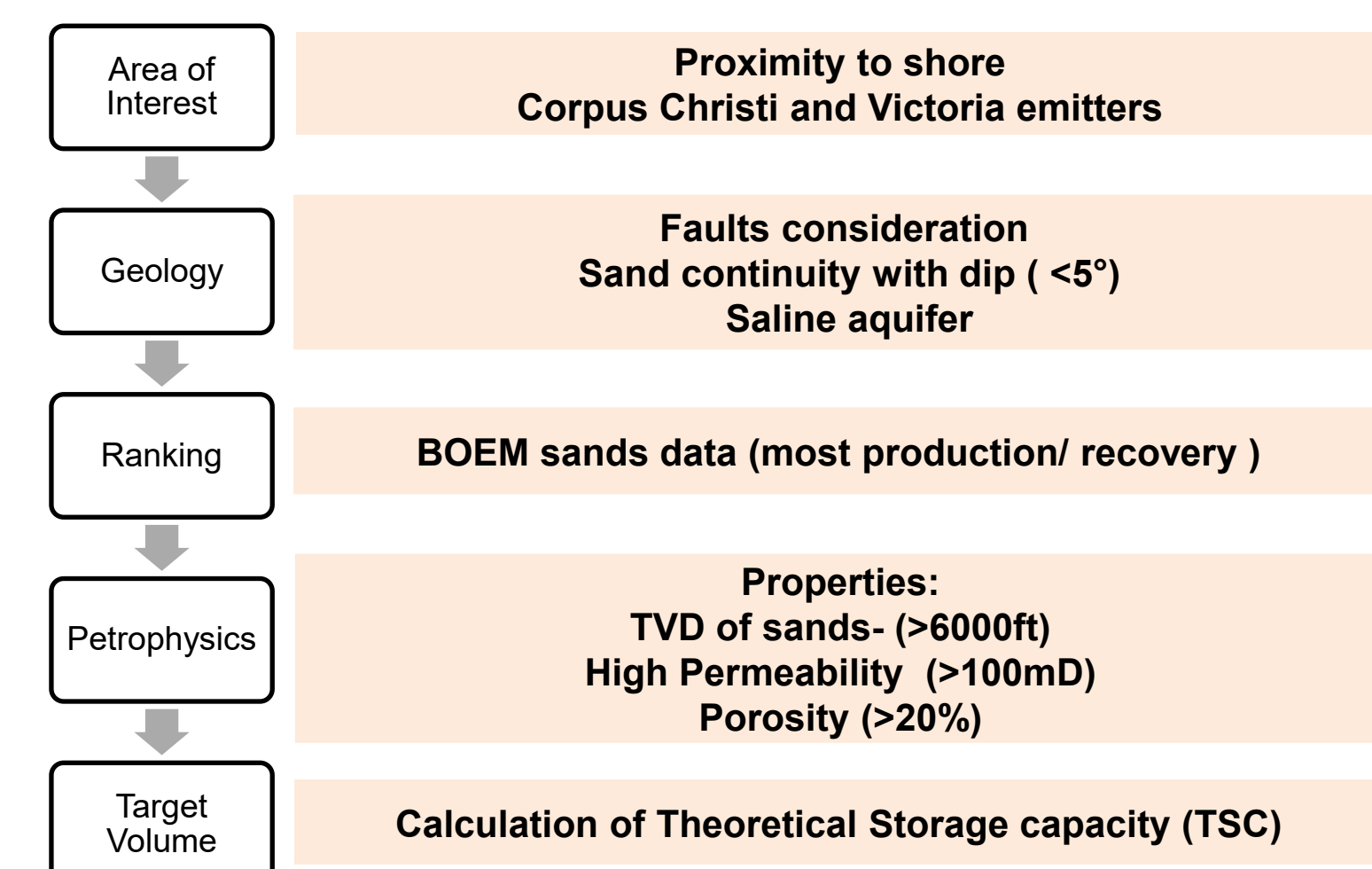
3 Characterization



5 Economics



2 Assessment - Overview



Reservoir Characterization:
L. Miocene, "Discorbia B" Fm.
~200 ft average thickness
0.27 average porosity
Containment:
700-1000 ft thick caprock, good lateral continuity
Injection:
2 Megaton/year combined for (two wells)

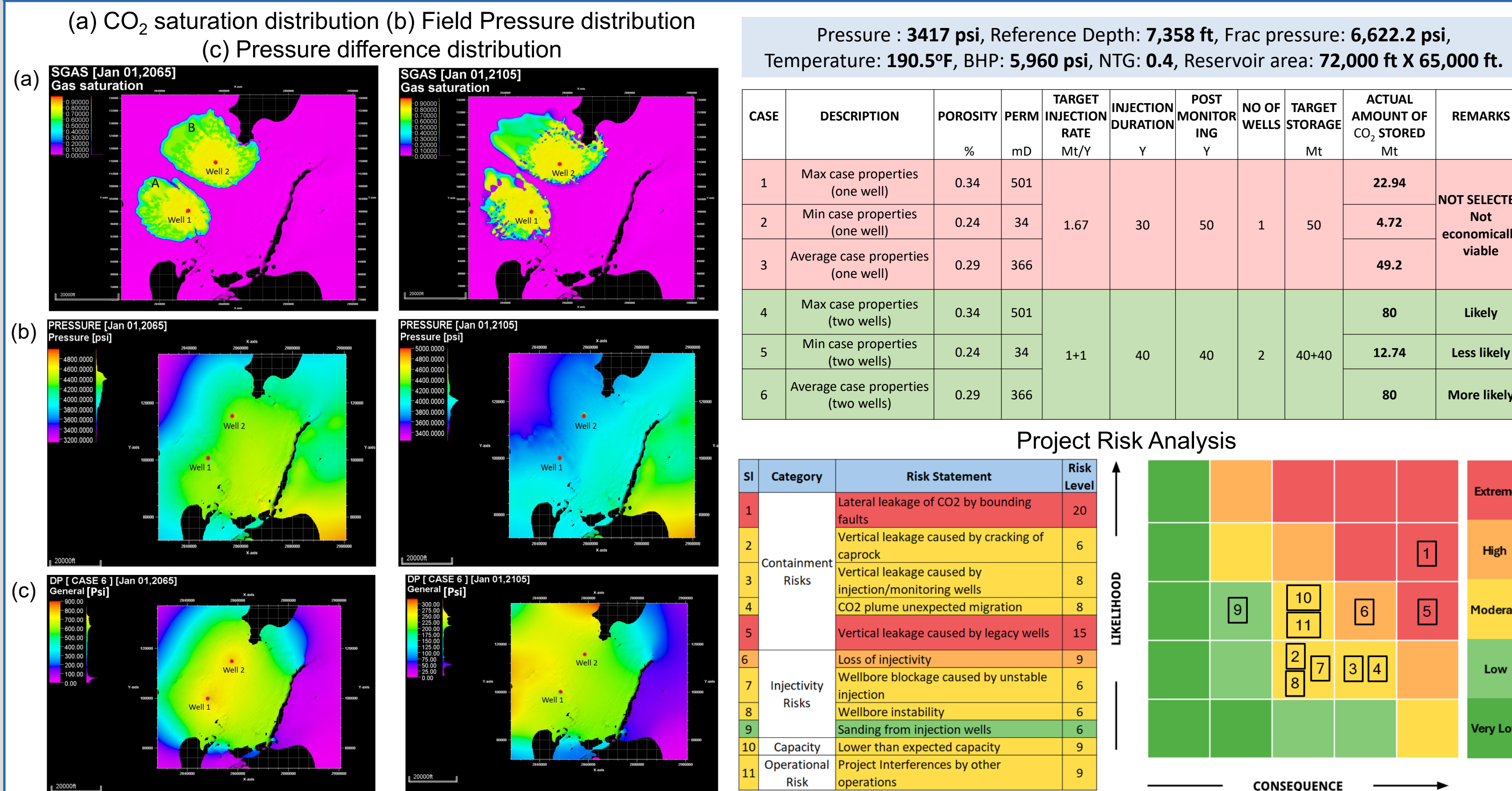
$$TSC = EUR * B_g * \rho_{CO_2} * E_f$$

$$TSC = EUR * B_o * \rho_{CO_2} * E_f$$

Assumed $\rho_{CO_2} = 0.7 \text{ tons/m}^3$

Where TSC is the Theoretical Storage Capacity; EUR is the estimated ultimate recovery; B_g is the formation volume factor for gas; B_o is the formation volume factor for oil; ρ_{CO_2} is the density of CO₂ and E_f is the efficiency factor.

4 Modeling, Simulation and MMV



6 Conclusions and Future Work

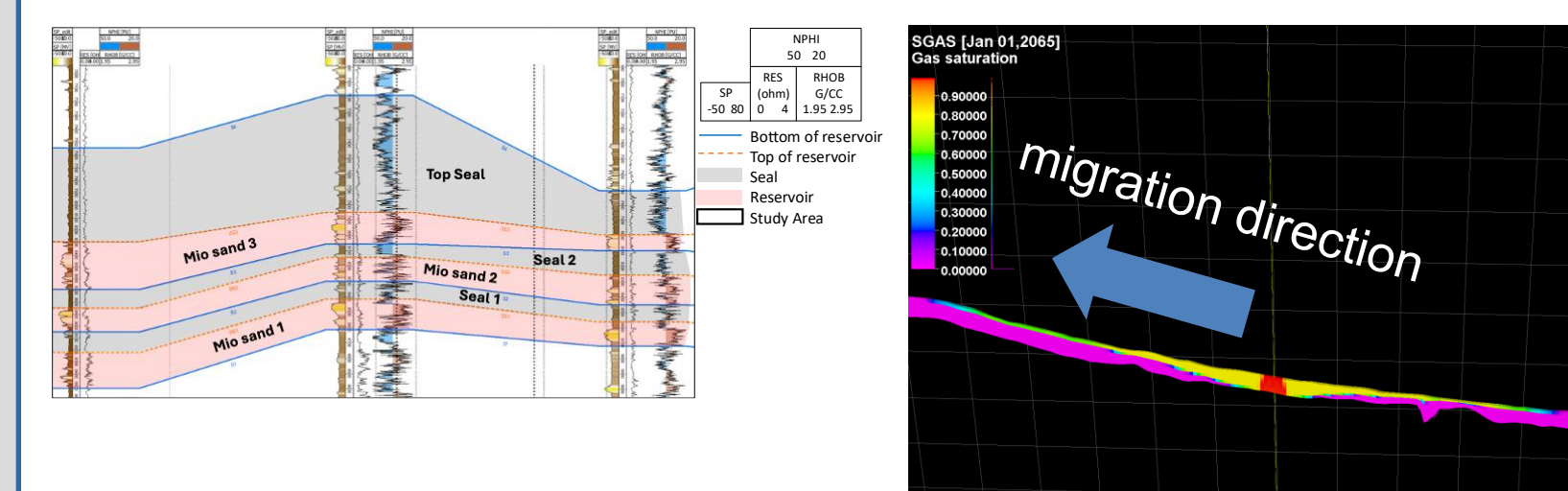
This project is capable of storing 80MMT of CO₂ over 40 years using 2 wells at injection rate of 1MT/Y for each well. The cost of project is \$5.5B and return on investment is 20.52%.

Primary project risks requiring monitoring and mitigation:

- Fluid pressure buildup along faults
- Legacy wells in storage interval & confining system

Future Work

- Stacked storage opportunities
- Additional injection well placement optimization
- Geomechanical model & analysis
- Continue updating business model under changing federal incentives
- Additional post-injection monitoring



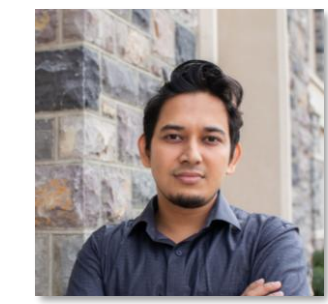
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Acknowledgments



Our SEG EVOLVE experience

"The SEG EVOLVE Program is a great opportunity to learn from industry datasets and mentors!"

